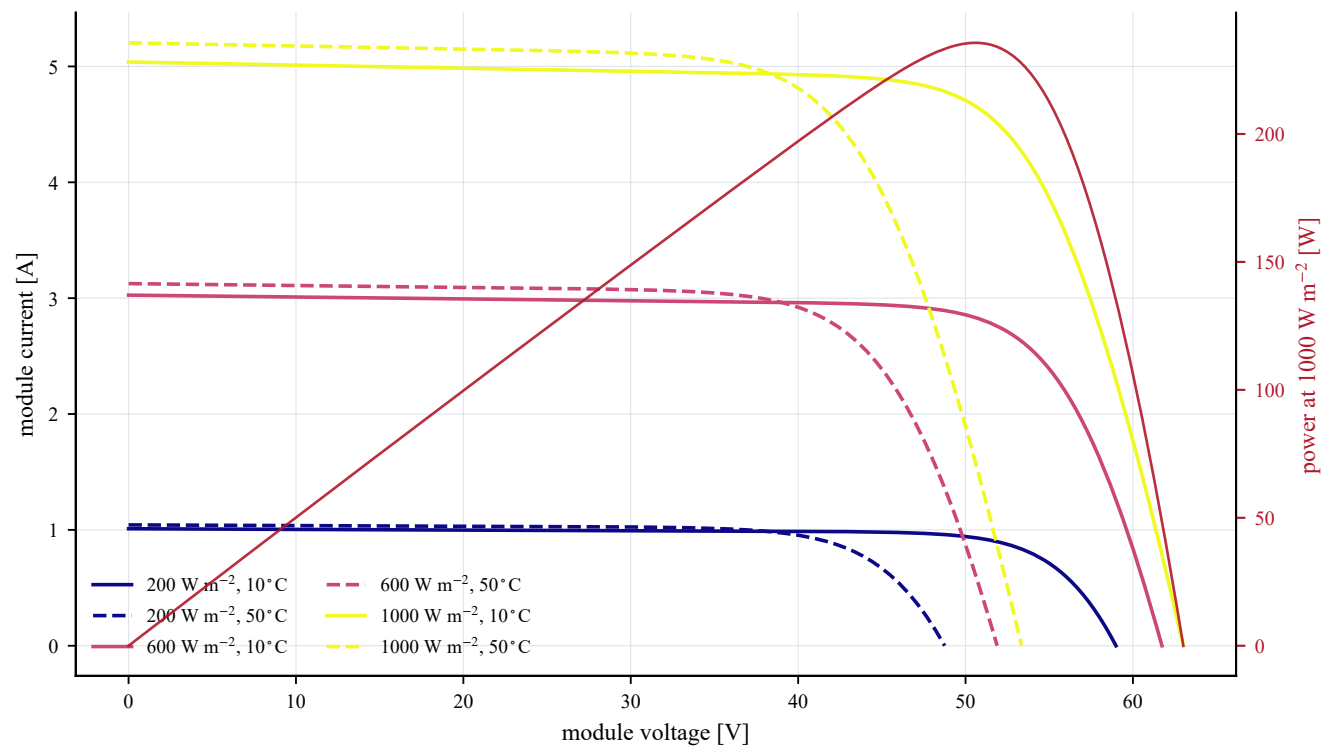
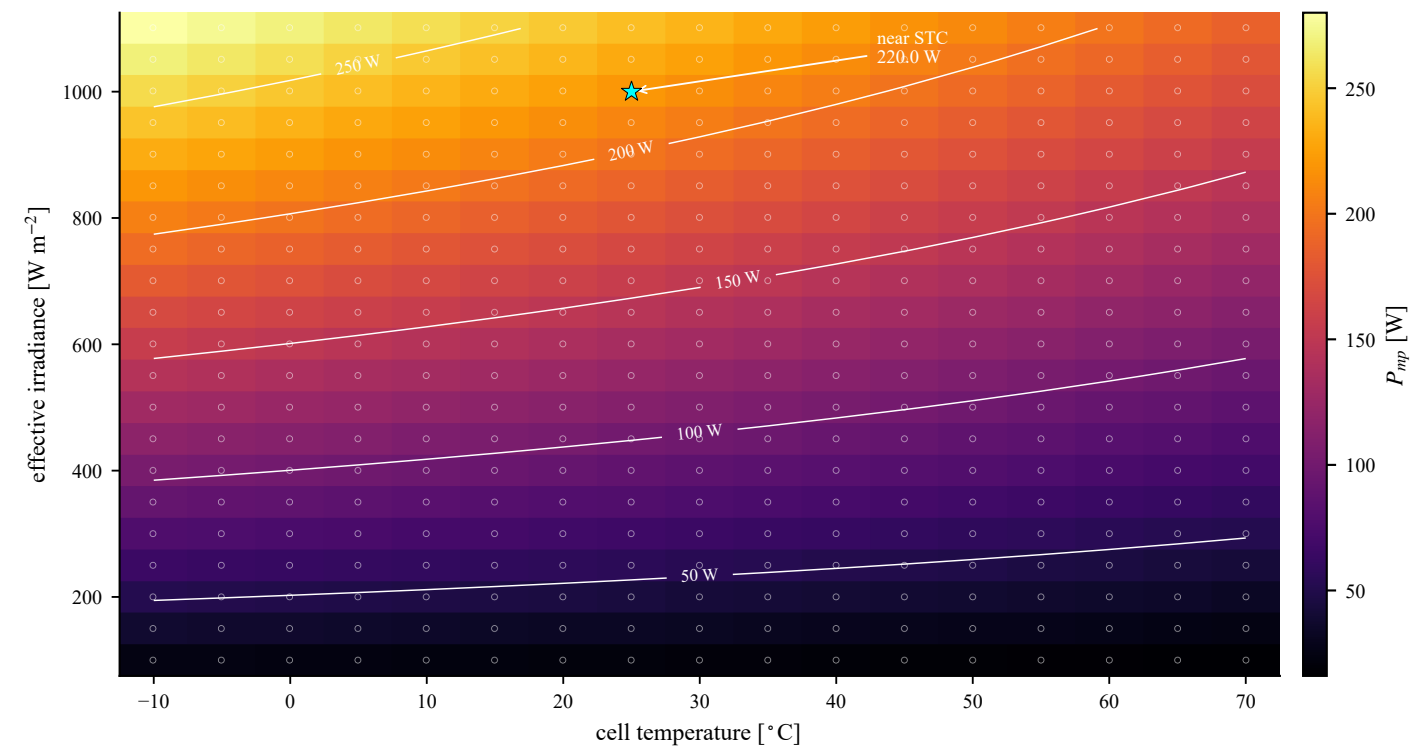


a

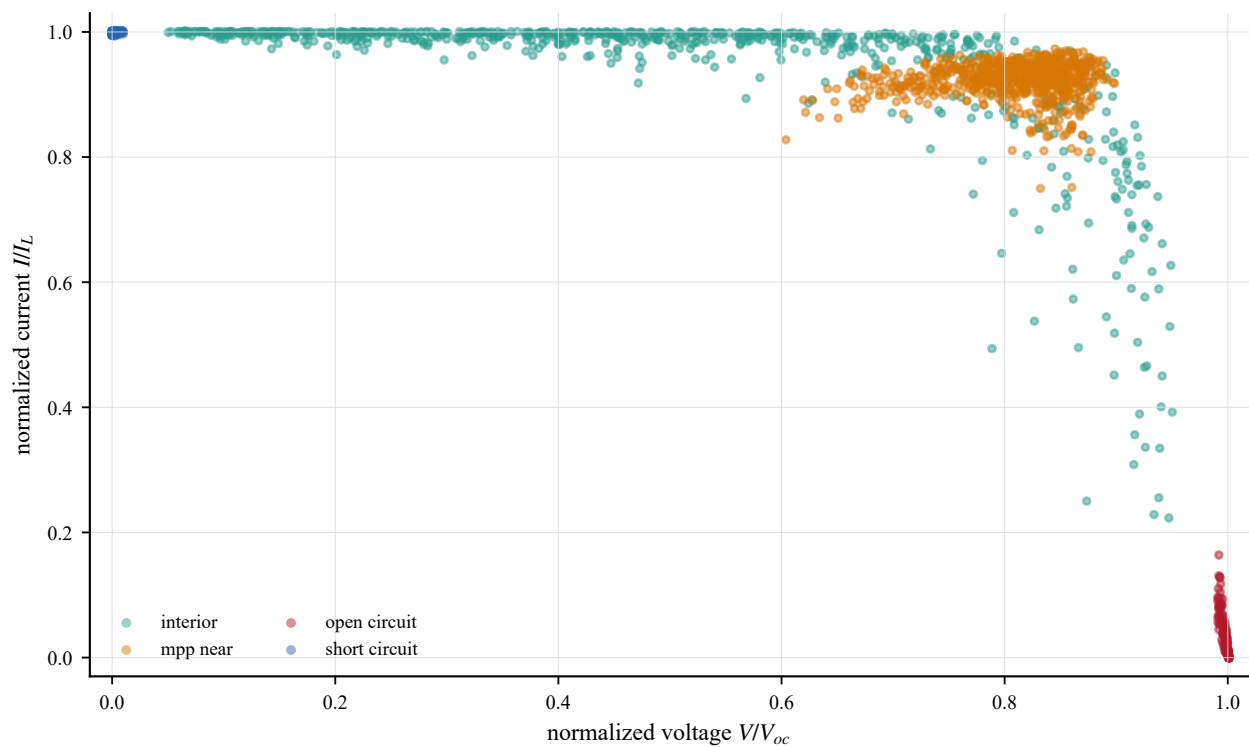
CEC module I-V and P-V curves from pvlb

**b**

Maximum-power surface: 357 independent operating points

**c**

Frozen benchmark coverage in normalized electrical coordinates

**d**

Independent pvlb Brent check against the 70-digit oracle

